SCHOOL HOMEWORK - 2022

CLASS: SERVER SECURITY

SUBJECT: HARDENING OF SELINUX AND WINDOWS SYSTEMS

FRÉDÉRIC PERRON

Table of Contents

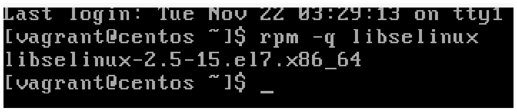
Part 1 : SELinux

Start a new VM (Linux CentOS/7) by using the given Vagrant file. Don't forget to put up to date the VM before starting the homework.

By using the CIS hardening guide by CentOS/7 (given document), complete the following tasks :

1. **Verify if SELinux is installed (control : 1.6.2). Provide a screenshot indicating whether SElinux is installed or not.**

Linux is indeed installed on my CentOS/7 machine. The command rpm -q libselinux was used. *See Image 1.*



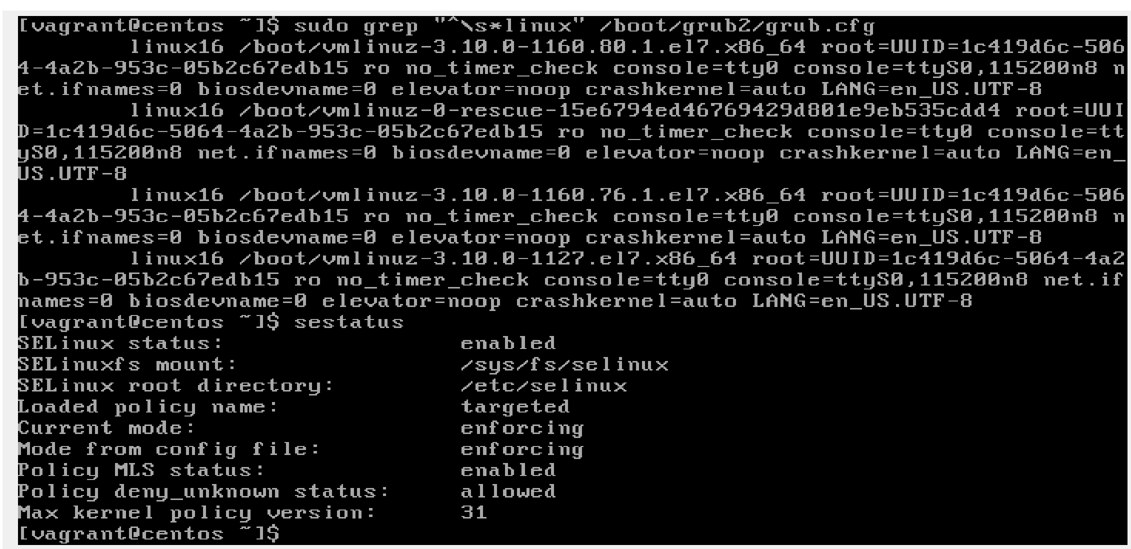
```
Last login: Tue Nov 22 03:29:13 on tty1
[vagrant@centos ~]$ rpm -q libselinux
libselinux-2.5-15.el7.x86_64
[vagrant@centos ~]$ _
```

Image 1

1. **Verify if SELinux isn't deactivated on the bootloader level (control 1.6.1.1).**

SELinux isn't deactivated at the bootloader level. The commands grep "^\s*linux" /boot/grub2/grub.cfg and sestatus were used.

a) **Provide a screenshot that shows if SELinux is deactivated at the bootloader level.**



```
[vagrant@centos ~]$ sudo grep "^\s*linux" /boot/grub2/grub.cfg
        linux16 /boot/vmlinuz-3.10.0-1160.80.1.el7.x86_64 root=UUID=1c419d6c-506
4-4a2b-953c-05b2c67edb15 ro no_timer_check console=tty0 console=ttyS0,115200n8 n
et.ifnames=0 biosdevname=0 elevator=noop crashkernel=auto LANG=en_US.UTF-8
        linux16 /boot/vmlinuz-0-rescue-15e6794ed46769429d801e9eb535cdd4 root=UUI
D=1c419d6c-5064-4a2b-953c-05b2c67edb15 ro no_timer_check console=tty0 console=tt
yS0,115200n8 net.ifnames=0 biosdevname=0 elevator=noop crashkernel=auto LANG=en_
US.UTF-8
        linux16 /boot/vmlinuz-3.10.0-1160.76.1.el7.x86_64 root=UUID=1c419d6c-506
4-4a2b-953c-05b2c67edb15 ro no_timer_check console=tty0 console=ttyS0,115200n8 n
et.ifnames=0 biosdevname=0 elevator=noop crashkernel=auto LANG=en_US.UTF-8
        linux16 /boot/vmlinuz-3.10.0-1127.el7.x86_64 root=UUID=1c419d6c-5064-4a2
b-953c-05b2c67edb15 ro no_timer_check console=tty0 console=ttyS0,115200n8 net.if
names=0 biosdevname=0 elevator=noop crashkernel=auto LANG=en_US.UTF-8
[vagrant@centos ~]$ sestatus
SELinux status:                 enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:             targeted
Current mode:                   enforcing
Mode from config file:          enforcing
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Max kernel policy version:      31
[vagrant@centos ~]$
```

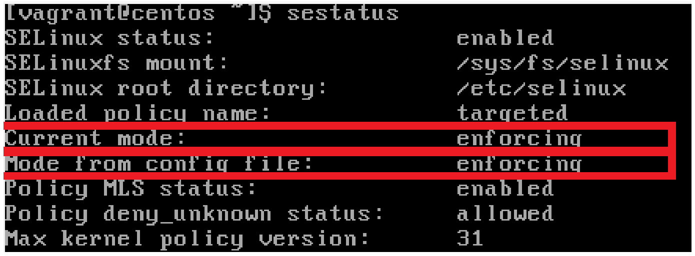
Image 2

a) **What is your conclusion? Is SELinux deactivated at the bootloader level?**

Linux isn't deactivated at the bootloader level as we have seen on the picture above.

2. **Verify the state of SELinux (Disabled, Permissive, Enforcing) (1.6.1.2). Provide a screenshot that shows the actual state of SELinux**

The state of SELinux is configured to Enforcing. The same command sestatus was used. See Image 3 below.

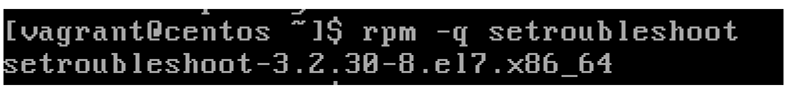


```
[vagrant@centos ~]$ sestatus
SELinux status:                 enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:             targeted
Current mode:                   enforcing
Mode from config file:          enforcing
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Max kernel policy version:      31
```

Image 3

3. **Verify if the SELinux debugging tool is installed. (control 1.6.1.4). Provide a screenshot that shows if the debugging tool is installed or not.**

The debugging tool is activated and the version is displayed. The command rpm -q setroubleshoot was used. *See image 4.*



```
[vagrant@centos ~]$ rpm -q setroubleshoot
setroubleshoot-3.2.30-8.el7.x86_64
```

Image 4

4. **Verify which policy is activated (targeted, minimum, mls) and change it. (control 1.6.1.3).**

The policy targeted is activated. It was changed to mls. *See Image 5 and 6 below.*

a) **Provide a screenshot that shows the activated policy in SELinux**

```
[vagrant@centos ~]$ sestatus
SELinux status:                 enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:             targeted
Current mode:                   enforcing
Mode from config file:          enforcing
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Max kernel policy version:      31
```

Image 5

b) **Provide a screenshot of SELinux after changing the policy.**

To change the policy from targeted to mls, the mls package first had to be installed using the command: sudo yum install selinux-policy-mls. Then, the vi text editor was opened with the following command: vi /etc/selinux/config.
Once inside the text editor, we were able to change the policy to mls. *See Image 6 to view the mls policy.*

```
"config" 12L, 536C written
[vagrant@centos selinux]$ sestatus
SELinux status:                 enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:             mls
Current mode:                   enforcing
Mode from config file:          permissive
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Max kernel policy version:      31
```

Image 6

Part 2 : Windows

Start a new VM (Windows Server 2012-R2) using the provided Vagrantfile. You may use the same VM as the one used in the course. Create a Group Policy Object (GPO) and name it GPO_TS. Link this GPO to the domain you have already created (e.g., CR345_domain.net).

Using the CIS Hardening Guide for Windows Server 2012-R2 (provided document), complete the following tasks on the created GPO:

1. **Check the policy related to password history (1.1.1 (L1) Ensure 'Enforce password history').**

   a) Provide a screenshot showing the current value of the password history policy.

Policy	Policy Setting
Enforce password history	24 passwords remembered
Maximum password age	42 days
Minimum password age	1 days
Minimum password length	7 characters
Password must meet complexity requirements	Enabled
Store passwords using reversible encryption	Disabled

   a) Modify the password history to ``5``. Provide a screenshot that shows the new value.

Policy	Policy Setting
Enforce password history	5 passwords remembered
Maximum password age	42 days
Minimum password age	1 days
Minimum password length	7 characters
Password must meet complexity requirements	Enabled
Store passwords using reversible encryption	Disabled

2. **Check the policy related to maximum password age (1.1.2 (L1) Ensure 'Maximum password age').**

The maximum age is set to 42 days.

a) **Provide a screenshot that shows the current maximum age of passwords.**

b) **Modify the maximum password age to 30 days. Provide a screenshot that shows the new value.**

3. **Check the policy related to minimum password length (1.1.4 (L1) Ensure 'Minimum password length').**

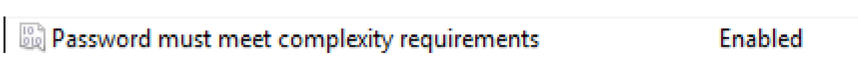
It is by default set to 7 characters.

a) **Provide a screenshot that shows the minimum length of passwords set right now.**

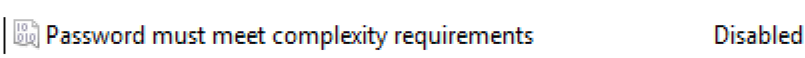
b) **Modify the length of passwords to 8 characters. Provide a screenshot that shows the new value.**

4. **Check the policy related to password complexity requirements (1.1.5 (L1) Ensure ‘Password must meet complexity requirements’)**

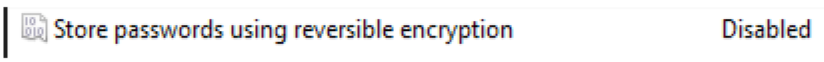
a) **Provide a screenshot that shows the status of the policy related to the complexity of passwords.**



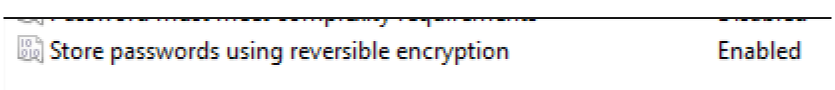
b) **Modify the status of the policy. Provide a screenshot that shows the new value.**



5. **Verify the policy related to password storage (1.1.6 (L1) Ensure 'Store passwords using reversible encryption').**
    a) **Provide a screenshot showing the status of the password storage policy.**



    b) **Change the policy status. Provide a screenshot showing the new value.**



References :

-https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/8/html/using_selinux/using-multi-level-security-mls_using-selinux#switching-the-selinux-policy-to-mls_using-multi-level-security-mls

-https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/6/html/security-enhanced_linux/enabling-mls-in-selinux

-Windows & CentOS CIS benchmark guides (given in class by the professor)

Setting : Must prepare a PDF with the 10 required questions and their respective answers

Evaluation : Each answer counts for 1 point, totalizing 17. Good luck!